Template Document

Academic Support Agent

February 19, 2025

: : February 19, 2025

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1

1.1

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:

0.4

[scale=0.75] (A)at (0,1); (B)at (3,0); (C)at (3,2.5); (D)at (0,3.5);

(E)at (2,2); (F)at (5,1); (G)at (5,3.5); (H)at (2,4.5);

[thick](A)–(B)–(C)–(D)–cycle; [thick](B)–(F)–(G)–(H)–(D) (G)–(C); [dashed](H)–(E)–(F) (E)–(A);

<3->[red,very thick]$((H)!-0.3!(G))$–$((G)!-0.3!(H))$node[right]$_{(D_2)}$; [blue,very thick]$((B)!-0.3!(C))$–$((C)!-0.3!(B))$node[right]$_{(D_1)}$;

<4->[blue](G)circle(2.5pt)node[above right]$_{A}$;

<5->[dashed,blue,very thick]$((F)!-0.3!(G))$–$((G)!-0.3!(F))$; [thick]$((G)!0.3cm!(H))$–+(0,-0.3)–$((G)!0.3cm!(F))$;

<7->[OliveGreen,very thick]$((H)!-0.3!(D))$–$((D)!-0.3!(H))$node[above]$_{(D_3)}$;

0.5

<6->

- A (D_1) (D_2) (D_1) (D_2) .
- (D_1) (D_3)

1.2

.

1. .
2. .

0.3

[scale=0.75] (A)at (0,0); (B)at (3,0); (C)at (3,3); (D)at (0,3);

(E)at (1,1); (F)at (4,1); (G)at (4,4); (H)at (1,4);

[thick](A)–(B)–(C)–(D)–cycle; [thick](B)–(F)–(G)–(H)–(D) (G)–(C); [dashed](H)–(E)–(F) (E)–(A);

<2->[red,very thick]$((H)!-0.3!(C))$node[above]$_{(\Delta_1)}$–$((C)!-0.3!(H))$; [red,very thick,dashed]$((E)!-0.3!(B))$node[above]$_{(\Delta_2)}$–$((B)!-0.3!(E))$; [red,very thick](B)–$((B)!-0.3!(E))$;

<3->[blue,very thick]$((G)!-0.2!(D))$node[above]$_{(D)}$–$((D)!-0.2!(G))$; [thick](2.2,3.4)–(2.4,3.45)–(2.2,3.55);

0.65

- <2->
  (Δ_1) (Δ_2)
- <3->
  (D) (Δ_1)
- <4->
  (D) (Δ_2)

2

2.1

. .

:

0.35
[scale=0.5] (A)at (0,0); (B)at (3,0); (C)at (3,3); (D)at (0,3);
(E)at (1,1); (F)at (4,1); (G)at (4,4); (H)at (1,4);
[thick](A)–(B)–(C)–(D)–cycle; [thick](B)–(F)–(G)–(H)–(D) (G)–(C); [dashed](H)–(E)–(F) (E)–(A);
<2-> ıinB,F[red](ı)circle(2.5pt)node[right,black]ß; [red](A)circle(2.5pt)node[left,black]A; [red,very thick]$((B)! - 0.2!(C))$–$((C)! - 0.3!(B))$node[below left](Δ); [thick](2.7,0)–(2.7,0.3)–(3,0.3);
[thick](3,0.3)–(3.2,0.55)–(3.2,0.2);
<6->ıin 0.2,0.4,...,2.8 $((E) + (ß, 0))$–$((A) + (ß, 0))$;
0.6 <5-> $(\Delta) : (BA)\ (BF)\ (ABF)$

2.2

.

1. <2-> .
2. <2-> .
3. <3-> .
4. <3-> .
5. <4-> .
6. <4-> .

3

3.1

.

.

0.35
[scale=0.75] (A)at (0,1); (B)at (3,0); (C)at (3,2.5); (D)at (0,3.5);
(E)at (2,2); (F)at (5,1); (G)at (5,3.5); (H)at (2,4.5);
<3>[red!30](C)–(B)–(F)–(G)–cycle; [green!30](C)–(G)–(H)–(D)–cycle;
<4->[red!30](C)–(B)–(F)–(G)–cycle; [gray!50](A)–(B)–(C)–(D)–cycle;
[thick](A)–(B)–(C)–(D)–cycle; [thick](B)–(F)–(G)–(H)–(D) (G)–(C); [dashed](H)–(E)–(F) (E)–(A);
ıinA,B,C,D,E,F,G,H[blue](ı)circle(2pt)node[above left]ß;
0.6

<2->
$ABCDEFGH$

- <3->
  (BCG) (DCG)
- <4->
  (BCG) (ADC)

3.2

.

1. .
2. (P) (P') (Q) (P) (P') (Q).

4

enumerate item[circle]
47 209 $ABCD$ $AD = DC$ $AB = BC$ M $[AC]$.

1. (AC) (BDM).
2. (AC) (BD).

0.27 [scale=0.7,equal/.style=fill=red,inner sep=0pt,minimum width=2mm] (D) at (0,0); (C) at (2.5,-1); (B) at (5,0); (A) at (2,4); (M) at ($(A)!0.5!(C)$);
<5->[red!20](M)–(D)–(B)–cycle;
[dashed](B)–(D); [thick](D)–(C)–(B)–(A)–cycle (A)–(C);
ı/in A/above,B/right,C/below,D/left[blue](ı)circle (2pt)node[]ß;
<4-> [red](M)circle(2pt)node[above right]M; [red,thick](B)–(M)–(D); ıin A,C [equal,rotate=5] at ($(M)!.5!(ß)$);
<3-> ıin .5,.52[equal,rotate=45] at ($(A)!ß!(B)$) ; ıin .5,.54[equal,rotate=100] at ($(B)!ß!(C)$) ; ıin .48,.5,.52[equal,rotate=-30] at ($(A)!ß!(D)$) ; ıin .47,.5,.53[equal,rotate=70] at ($(D)!ß!(C)$) ;
0.73

- <6-| alert@6>
  ABC ADC
  (BM) (DM) .
  (AC)
  (BDM).
- <7-| alert@7>
  (AC) (BDM) (BD).

<8->